

Coordinates on the plane; the midpoint of a segment

Two numbers fix a point — and from there geometry can be done with arithmetic.

LESSON 33–34

2 × 40 MIN

GEOMETRY 8, TEMA 31

STAGE 9 · 3.1

LESSON CLOCK

6'

10'

12'

8'

4'

Plot five points from dictation	6 min
Reading the plane	10 min
The midpoint formula	12 min
Problems both ways	8 min
Homework	4 min

LEARNING OBJECTIVES

- Plot and read points in all four quadrants.
- Find the midpoint of a segment from the coordinates of its ends.
- Find an endpoint when the midpoint and the other end are known.
- Use coordinates to show that a quadrilateral is a parallelogram.

TEXTBOOK

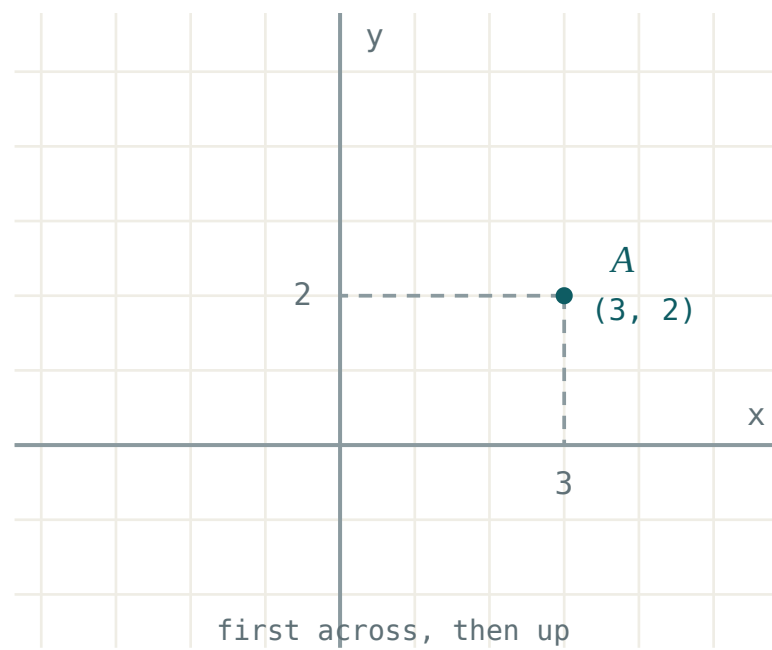
National	Tema 31, pp. 69–71
Cambridge	Learner’s Book pp. 56–60
Workbook	Workbook 3.1

01

Explanation

A point is a pair of numbers

Two perpendicular axes meet at the **origin** $O(0; 0)$. Any point of the plane is then fixed by an **ordered pair** $A(x; y)$: first go x along, then y up.



A point is fixed by two numbers — how far across, then how far up. The order matters.

The order matters absolutely: (3; 5) and (5; 3) are different points. The axes cut the plane into four **quadrants**:

QUADRANT	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
x	+	−	−	+
y	+	+	−	−

A point on the *x*-axis has *y* = 0; a point on the *y*-axis has *x* = 0. The origin is the only point on both.

The midpoint

MIDPOINT OF A SEGMENT

If $A(x_1; y_1)$ and $B(x_2; y_2)$, the midpoint M of AB is

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

— the average of the two *x*’s and the average of the two *y*’s.

Why the average? Going from *A* to *B* the *x*-coordinate changes by $x_2 - x_1$; halfway there it has changed by half of that:

$$x_1 + \frac{x_2 - x_1}{2} = \frac{2x_1 + x_2 - x_1}{2} = \frac{x_1 + x_2}{2}$$

The same argument works for y . So the formula is not something to memorise blindly — it says “stand halfway”.

READ THE QUESTION

“Find the midpoint” means average. “Find the other end, given the midpoint” means the reverse: $x_2 = 2x_M - x_1$. Doubling and subtracting, not averaging.

Why coordinates help

Once points are numbers, statements about shapes become statements about arithmetic. Take $A(1; 2)$, $B(5; 3)$, $C(6; 6)$, $D(2; 5)$. Is $ABCD$ a parallelogram?

midpoint of AC : $(3.5; 4)$ | midpoint of BD : $(3.5; 4)$

The diagonals share a midpoint, so they bisect each other — and by the test proved in Quarter I, $ABCD$ is a parallelogram. No drawing, no measuring, no doubt.

This is the **coordinate method**: translate the geometry into coordinates, do the algebra, translate the answer back. The rest of this chapter is built on it.

02 Worked examples**EXAMPLE 1** Find the midpoint of $A(-3; 4)$ and $B(7; -2)$.

$$\textcircled{1} \quad x_M = \frac{-3 + 7}{2} = 2$$

Average the x-coordinates.

$$\textcircled{2} \quad y_M = \frac{4 + (-2)}{2} = 1$$

Average the y-coordinates.

$$\textcircled{3} \quad M(2; 1)$$

Check: it lies between the two, as it must.

ANSWER $M(2; 1)$

EXAMPLE 2 $M(4; -1)$ is the midpoint of AB and $A(1; 3)$. Find B .

$$\textcircled{1} \quad \frac{1+x}{2} = 4$$

Write the midpoint formula and solve for x .

$$\textcircled{2} \quad 1+x=8, x=7$$

Double, then subtract.

$$\textcircled{3} \quad \frac{3+y}{2} = -1$$

$$\textcircled{4} \quad 3+y=-2, y=-5$$

$$\textcircled{5} \quad B(7; -5)$$

Check the midpoint of $(1;3)$ and $(7;-5)$ is $(4;-1)$ ✓

ANSWER

$B(7; -5)$

EXAMPLE 3 Show that $A(0; 0)$, $B(4; 1)$, $C(5; 5)$, $D(1; 4)$ form a parallelogram.

- 1 Find the midpoint of each diagonal.
The bisection test.

- 2 $AC: (2.5; 2.5)$

- 3 $BD: (2.5; 2.5)$
The same point.

- 4 The diagonals bisect each other, so $ABCD$ is a parallelogram.
Test proved in Quarter I.

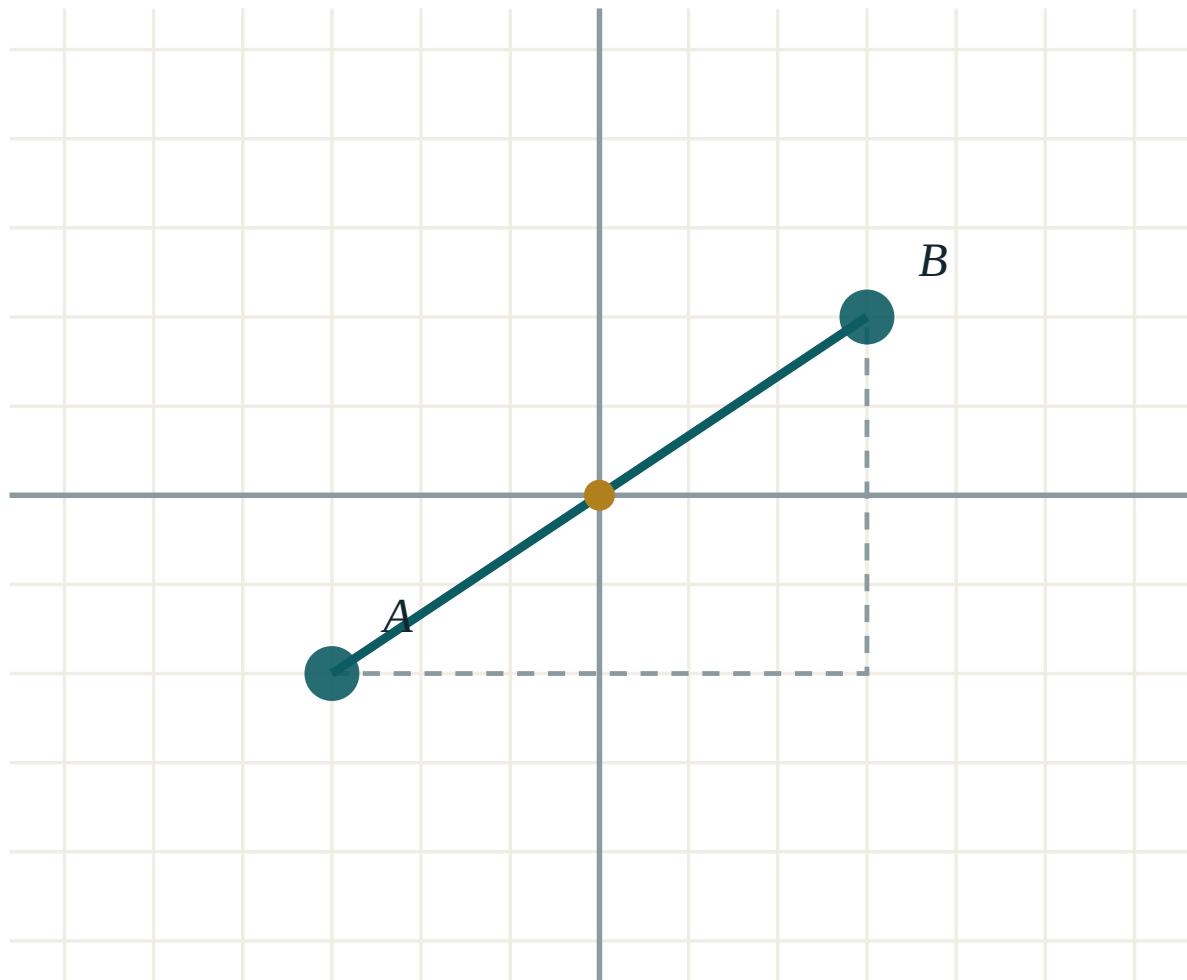
ANSWER

Yes — the diagonals share the midpoint $(2.5; 2.5)$

03 Interactive model

Drag A and B; the golden dot is always the midpoint, and its coordinates are the two averages.

Two points and their midpoint



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinate plane	Koordinatalar tekisligi	Координатная плоскость
Origin	Koordinata boshi	Начало координат
Abscissa (x-coordinate)	Abssissa	Абсцисса
Ordinate (y-coordinate)	Ordinata	Ордината
Quadrant	Chorak	Координатная четверть
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Midpoint	O'rta nuqta	Середина отрезка
Segment	Kesma	Отрезок
Axis	O'q	Ось
Coordinate method	Koordinatalar usuli	Метод координат

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The midpoint of $(2; 6)$ and $(8; 2)$ is:

$(5; 4)$

$(10; 8)$

$(6; 4)$

$(3; 2)$

$(-4; 3)$ lies in quadrant:

I

II

III

IV

If $M(3; 5)$ is the midpoint of $A(1; 2)B$, then B is:

$(2; 3.5)$

$(5; 8)$

$(4; 7)$

$(6; 10)$

A point on the y -axis has:

$$y = 0$$

$$x = 0$$

$$x = y$$

$$x = 1$$

The midpoint of a segment from $(-5; -3)$ to $(5; 3)$ is:

$$(0; 0)$$

$$(5; 3)$$

$$(0; 3)$$

$$(10; 6)$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the midpoint of $(0; 0)$ and $(6; 8)$.

ANSWER $(3; 4)$

2. Find the midpoint of $(2; 5)$ and $(8; 11)$.

ANSWER $(5; 8)$

3. In which quadrant is $(3; -7)$?

ANSWER *IV*

4. Find the midpoint of $(-2; 4)$ and $(6; 4)$.

ANSWER $(2; 4)$

5. Write the coordinates of the origin.

ANSWER $(0; 0)$

6. Find the midpoint of $(-1; -1)$ and $(5; 7)$.

ANSWER $(2; 3)$

7. A point has $x = 0$ and $y = -4$. Where does it lie?

ANSWER On the y -axis, below the origin.

Medium · the standard question

7 PROBLEMS

1. Find the midpoint of $(-7; 2)$ and $(3; -8)$.

ANSWER $(-2; -3)$

2. $M(2; 3)$ is the midpoint of $A(-1; 5)B$. Find B .

ANSWER $B(5; 1)$

3. $M(0; 0)$ is the midpoint of $A(-4; 6)B$. Find B .

ANSWER $B(4; -6)$

4. Find the midpoint of $(1.5; 2.5)$ and $(4.5; 7.5)$.

ANSWER $(3; 5)$

5. The midpoints of the diagonals of $ABCD$ are $(2; 3)$ and $(2; 3)$. What can you conclude?

ANSWER The diagonals bisect each other, so $ABCD$ is a parallelogram.

6. $A(1; 1)$, $B(7; 3)$. Find the point one quarter of the way from A to B .

ANSWER $(2.5; 1.5)$ — the midpoint of A and the midpoint of AB

7. Find the midpoint of the segment joining $(a; b)$ and $(3a; 5b)$.

ANSWER $(2a; 3b)$

Hard · stretch and reason

7 PROBLEMS

1. $A(1; 2)$, $B(5; 4)$, $C(7; 8)$. Find D so that $ABCD$ is a parallelogram.

ANSWER $D(3; 6)$ — the diagonals must share the midpoint $(4; 5)$

2. Show that $(0; 0)$, $(6; 2)$, $(8; 8)$, $(2; 6)$ form a parallelogram.

ANSWER Both diagonals have midpoint $(4; 4)$.

3. The vertices of a triangle are $(0; 0)$, $(6; 0)$, $(2; 8)$. Find the midpoints of all three sides.

ANSWER $(3; 0)$, $(4; 4)$, $(1; 4)$

4. M is the midpoint of AB , with $A(-3; 7)$ and $M(1; 1)$. Find B and the midpoint of AM .

ANSWER $B(5; -5)$, midpoint of AM is $(-1; 4)$

5. $P(2; 3)$ and $Q(10; 7)$. Find the two points that divide PQ into three equal parts.

ANSWER $(4\frac{2}{3}; 4\frac{1}{3})$ and $(7\frac{1}{3}; 5\frac{2}{3})$

6. Prove that the diagonals of the quadrilateral $(0;0)$, $(a;0)$, $(a+b;c)$, $(b;c)$ bisect each other.

ANSWER Both diagonals have midpoint $(\frac{a+b}{2}; \frac{c}{2})$, so the figure is a parallelogram for all a , b , c .

7. $A(1; 1)$ and $B(9; 5)$. A point C on AB has $AC = 3 \cdot CB$. Find C .

ANSWER $(7; 4)$

07 Homework

Homework — 6 problems

Geometry 8, Tema 31, pp. 69–71. Sketch the axes for every problem, even the easy ones.

- Find the midpoint of $(3; 7)$ and $(11; 1)$.
- Find the midpoint of $(-6; 2)$ and $(4; -8)$.
- $M(3; -2)$ is the midpoint of $A(7; 4)B$. Find B .
- Plot $A(-2; 3)$, $B(4; 3)$, $C(4; -1)$, $D(-2; -1)$ and name the shape.

5. *Show that $(1; 1)$, $(5; 2)$, $(6; 6)$, $(2; 5)$ form a parallelogram.*
6. *Find the midpoints of all three sides of the triangle $(0; 0)$, $(8; 0)$, $(4; 6)$.*